Inequalities Notation

eg set of all real numbers between 2 and 6

including 2 and 6	excluding 2 and 6	
$[2,6]$	$(2,6)$	$\{2,6\}$

only 2 and 6.

$[6]$ ✗

$[5,5]$ ✗

$(2,2)$ ✗

$[8,3]$ ✗

$(8,3]$ ✗

Q $A=[2,6)$, $B=[5,10)\cup[14,18]$

$A\cup B$, $A\cap B$

M1

$[2,10)$ | $[5,6)$

M2

$A\cap B=[5,6)$, $A\cup B=[2,10)$

open bracket

$(\ )$

closed bracket

$[\ ]$

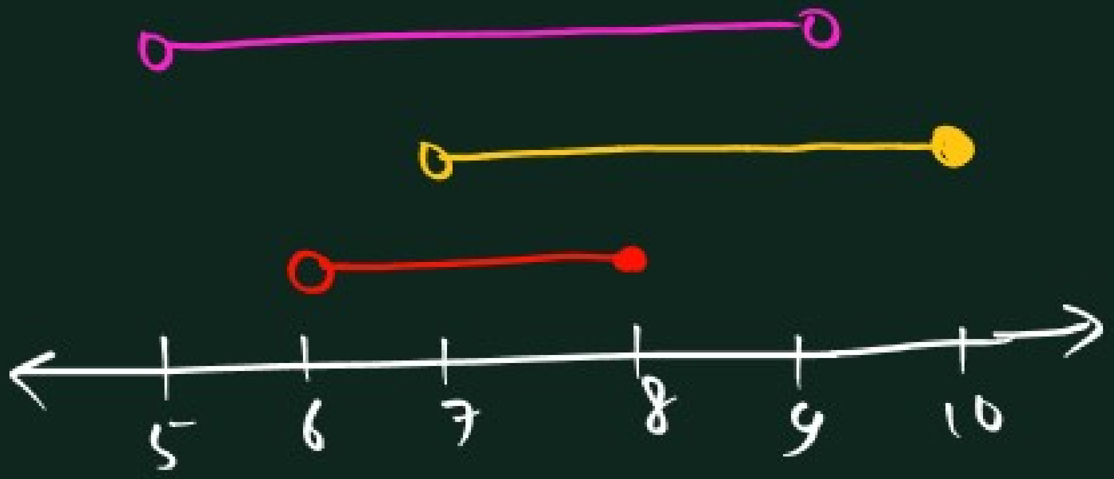
include ●

exclude ○

Q1 $A = (6, 8]$, $B = (7, 10]$, $C = (5, 9)$, find $A \cup B \cup C$ and $A \cap B \cap C$

Q2 $A = [2, 6) \cup [8, 10)$, $B = (4, 8]$, find $A \cup B$ and $A \cap B$.

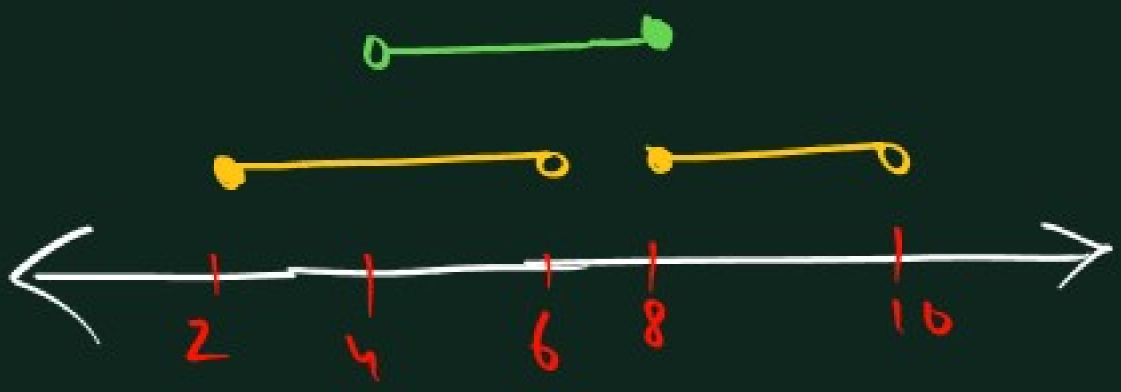
Soln 1



$A \cap B \cap C = (7, 8]$

$A \cup B \cup C = (5, 10]$

Soln 2



$A \cap B \cap C = (4, 6) \cup \{8\}$

$A \cup B \cup C = [2, 10)$

Properties of inequality

$a < b$

a is less than b

$x \geq 3$

x is greater than equal to 3

$x > 0$

x is +ve

$x \leq 6$

$x \in (-\infty, 6]$

$3 \leq x < 7$

$x \in [3, 7)$

→ addition or subtraction by a fixed number won't change sign of inequality ie transposition of add/subtract can be without changing sign.

→ multiplication or division by a fixed number will change the sign of inequality if it is a negative number. sign will change during transposition also, if it is a negative number.

$\rightsquigarrow$ cross is allowed only when denominators are positive [side of Nr does not change]

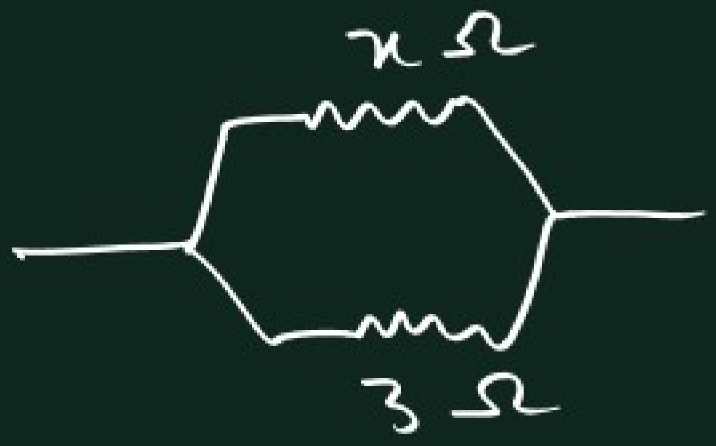
Q find x.

(i) $2x+3 \geq 3x-4$

(ii) $\frac{x+2}{3} < \frac{3x-5}{-2}$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Q



find x, if R_{eq} is more than 1 and less than 2.

(ii) $\dfrac{x+2}{3} < \dfrac{3x-5}{-2}$

$$\frac{x+2}{3} - \frac{3x-5}{-2} < 0$$

$$\frac{x+2}{3} + \frac{3x-5}{2} < 0$$

$$\frac{2(x+2)+3(3x-5)}{6} < 0$$

$$\frac{11x-11}{6} < 0$$

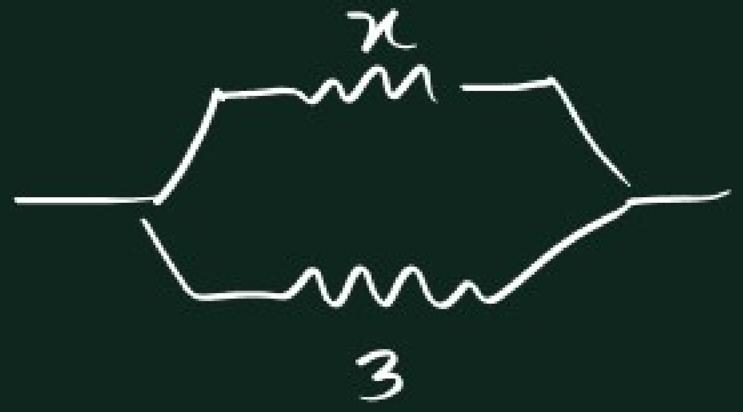
$$11x-11 < 0$$

$$11x < 11$$

$$x < 1$$

$$x \in (-\infty, 1)$$

Soln



$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{3x}{x+3}$$

$R_{eq} > 1$ and $R_{eq} < 2$

$$\frac{3x}{x+3} > 1 \qquad \frac{3x}{x+3} < 2$$

$$3x > x+3 \qquad 3x < 2(x+3)$$

$$2x > 3 \qquad 3x < 2x+6$$

$$x > \frac{3}{2} \qquad x < 6$$

$$\frac{3}{2} < x < 6$$

$$x \in \left(\frac{3}{2}, 6\right)$$

wavy curve method

$x^2 > 16$	$(x-4)(x-9) < 0$	$x^2 < -5$	$x^2 > -4$
$x \in (-\infty,-4) \cup (4,\infty)$	$x \in (4,9)$	$x \in \phi$	$x \in (-\infty,\infty)$
			$x \in R$

steps

1. make right side 0.
2. make coeff of the highest degree term (+ve)
3. factorise. [if already factorised then make coeff of highest degree term +ve in every bracket)
4. put roots in number line in order, do not put any unneccessary point.
5. make the extreme right interval (+ve). Now about the root 'α',

   if α is appearing odd times $\rightarrow$ sign changes about 'α'

   if α is appearing even times $\rightarrow$ " doesnot change about 'α'.

(6) If <0, then take negative portion

≤ 0, " " " " along with roots of Nr.

>0, " " positive portion

≥ 0, " " " " along with roots of Nr.

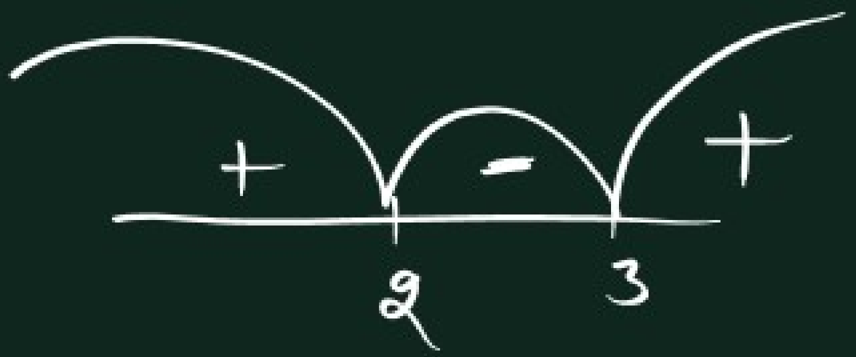
eg:

$$5x-6 \leq x^2$$

$$-x^2+5x-6 \leq 0$$

$$x^2-5x+6 \geq 0$$

$$(x-2)(x-3) \geq 0$$



$$x \in (-\infty, 2] \cup [3, \infty)$$

find x

(i) $x^2-4x+3 \leq 0$

(ii) $2x^2 < x$

(iii) $5x+6 \leq x^2$

(iv) $x^2+3x-5 > 0$

(i) $x^2-4x+3 \le 0$

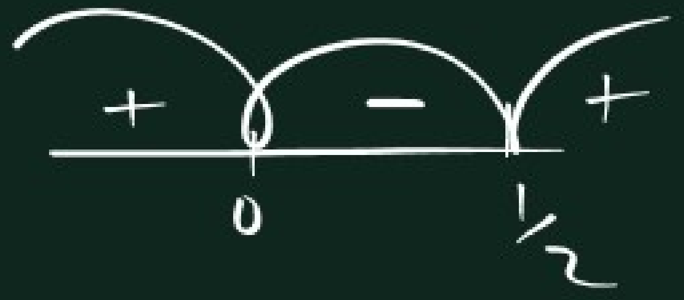
$(x-1)(x-3) \le 0$

$x \in [1, 3]$

(ii) $2x^2 < x$

$2x^2 - x < 0$

$x(2x-1) < 0$



$x \in \left(0, \frac{1}{2}\right)$

(iii) $5x+6 \le x^2$

$-x^2+5x+6 \le 0$

$x^2-5x-6 \ge 0$

$(x-6)(x+1) \ge 0$

$(-\infty, -1] \cup [6, \infty)$

(i) $x^2-2x+1 > 0$

(ii) $x^2+8x+16 < 0$

(iii) $x^2-6x+9 \ge 0$

(iv) $x^2+4x+4 \le 0$

$x^2+3x-5>0$

$$x=\frac{-3\pm\sqrt{9-4.1.(-5)}}{2\times 1}$$

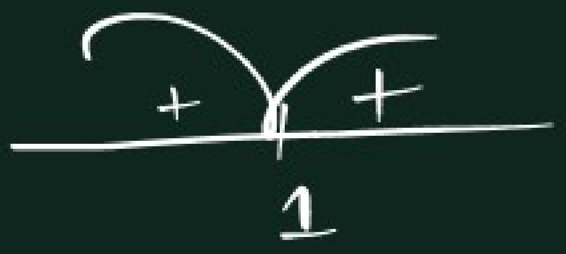
$$=\frac{-3\pm\sqrt{29}}{2}$$

$\frac{-3+\sqrt{29}}{2}$, $\frac{-3-\sqrt{29}}{2}$

$$x\in\left(-\infty,\frac{-3-\sqrt{29}}{2}\right)\cup\left(\frac{-3+\sqrt{29}}{2},\infty\right)$$

$x^2-2x+1>0$

$\Rightarrow (x-1)^2>0$



$(-\infty,1)\cup(1,\infty)$

$=(-\infty,\infty)-\{1\}$

$=R-\{1\}$

$x^2+8x+16<0$

$(x+4)^2<0$

$x\in\phi$

$x^2-6x+9\geq 0$

$(x-3)^2\geq 0$

$(-\infty,3]\cup[3,\infty)$

$(-\infty,\infty)$

$x\in R$

$x^2+4x+4\leq 0$

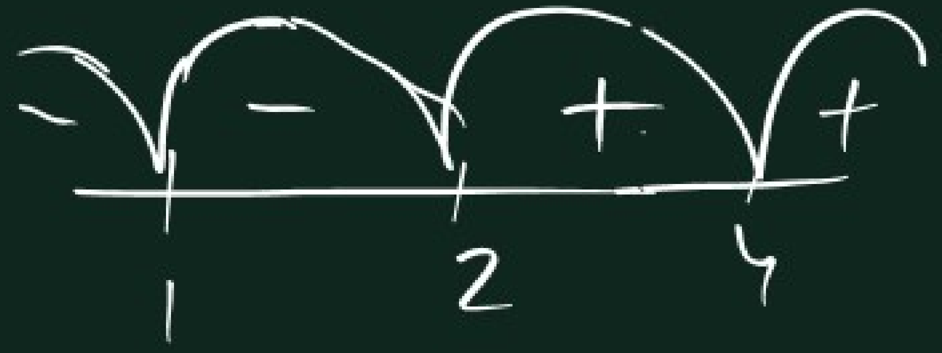
$(x+2)^2\leq 0$

$x\in\{-2\}$

(i) $\dfrac{(x-1)^2(2-x)^3}{(4-x)^2} \le 0$

$\Rightarrow \dfrac{-(x-1)^2(x-2)^3}{(x-4)^2} \le 0$

$\Rightarrow \dfrac{(x-1)^2(x-2)^3}{(x-4)^2} \ge 0$



$[2,4) \cup (4,\infty)$

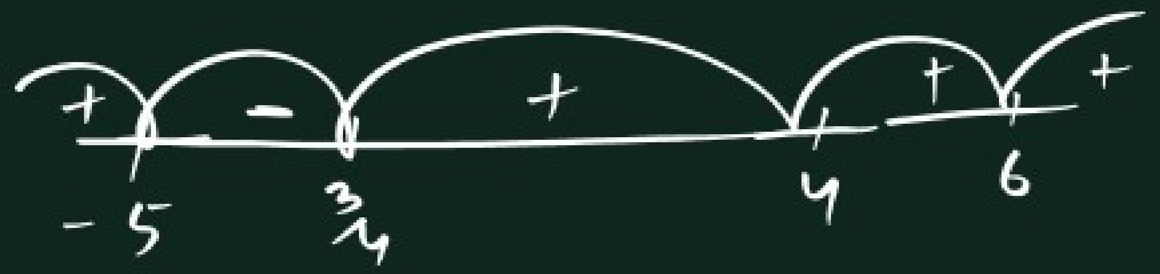
$[2,\infty) - \{4\}$

Q $\dfrac{x^2-6x+8}{x^2-8x+15} < 0$

Q $\dfrac{(4x-3)(x-4)^2}{(x+5)^3(x-6)^4} \ge 0$

Q $\dfrac{x}{x+2} \le \dfrac{1}{x}$

$$\frac{(4x-3)(x-4)^2}{(x+5)^3(x-6)^4} \geq 0$$



$$(-\infty,-5)\cup\left[\tfrac{3}{4},4\right]\cup[4,6)\cup(6,\infty)$$

$$x\in(-\infty,-5)\cup\left[\tfrac{3}{4},\infty\right)-\{6\}$$

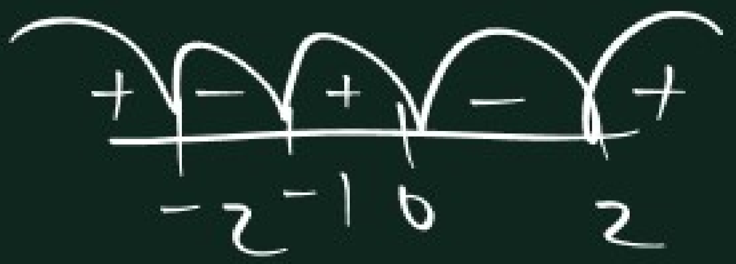
$$\frac{x}{x+2}\leq\frac{1}{x}$$

$$\frac{x}{x+2}-\frac{1}{x}\leq 0$$

$$\frac{x^2-(x+2)}{x(x+2)}\leq 0$$

$$\frac{x^2-x-2}{x(x+2)}\leq 0$$

$$\frac{(x-2)(x+1)}{x(x+2)}\leq 0$$



$$x\in(-2,-1]\cup(0,2]$$

$\rightsquigarrow ax^2+bx+c \longrightarrow$ always +ve quantity.

$a>0\ ,\ D<0$

$\rightarrow$ always positive factor can be ignored from inequality.

eg $(x^2-x-2)\cancel{(x^2-x+2)}<0$

$D=1+8=9$ $\qquad$ $D=1-8$
$=-7$

$(x^2-x-2)<0$

$(x-2)(x+1)<0$ $\qquad$ $+\ \ -\ \ +$ (-1, 2)

$x\in(-1,2)$

$x^4+x^2+5 \rightarrow$ a.p.

$2^x+1 \rightarrow$ a.p

$3^x \rightarrow$ a.p

Q $\dfrac{(x-1)(x^2+x+1)}{(x^2+1)(x+3)^2}\le 0$

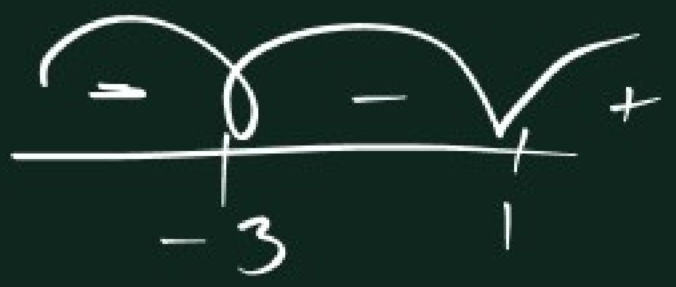
Q $x^2+x+1>0$

Q $(x^2-2x)(2x-2)-\dfrac{9(2x-2)}{x^2-2x}\le 0$

$D = -3$

$$\frac{(x-1)\,\cancel{(x^2+x+1)}}{\cancel{(x^2+1)}\,(x+3)^2} \le 0$$

$$\frac{(x-1)}{(x+3)^2} \le 0$$



$(-\infty,-3) \cup (-3,1]$

$(-\infty,1] - \{-3\}$

$x^2+x+1 > 0$

$1 > 0$

always true statement

$x \in (-\infty,\infty)$

$x \in R$

$$(x^2-2x)(2x-2) - \frac{9(2x-2)}{x^2-2x} \le 0$$

$$\frac{(x^2-2x)^2(2x-2) - 9(2x-2)}{x^2-2x} \le 0$$

$$\frac{(2x-2)\left[(x^2-2x)^2 - 3^2\right]}{x^2-2x} \le 0$$

$$\frac{2(x-1)\cancel{(x^2-2x+3)}(x^2-2x-3)}{x(x-2)} \le 0$$

$D=-8$

$D = 4+8$
$= 12$

$$\frac{2(x-1)(x-3)(x+1)}{x(x-2)} \le 0$$

$$\frac{(x-1)(x-3)(x+1)}{x(x-2)} \le 0$$

$-$ $+$ $-$ $+$ $-$ $+$

-1 $\quad 0$ $\quad 1$ $\quad 2$ $\quad 3$

$$(-\infty,-1] \cup (0,1] \cup (2,3]$$

Q

$$(x^2 - x - 1)(x^2 - x - 7) < -5$$

$x^2 - x = t$

$$(t-1)(t-7) < -5$$

$$t^2 - 8t + 7 < -5$$

$$t^2 - 8t + 12 < 0$$

$$(t-2)(t-6) < 0$$

$$(x^2 - x - 2)(x^2 - x - 6) < 0$$